

Data Analysis and Demand Prediction

A CAPSTONE PROJECT REPORT

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IN

Computer Science and Engineering

Submitted by

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DECLARATION

We, **Deepak. R (192511008.) Vijay. A (192511030)** of the **Computer Science and Engineering**, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled '**Data Analysis And Demand Prediction**' is the result of our own bonafide efforts. To the best of our knowledge, the work presented here in is original, accurate, and has been carried out in accordance with principles of engineering ethics.

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**Data Analysis And Demand Prediction**” has been carried out by **Deepak. R (192511008.) Vijay. A (192511030)** under the supervision of **Dr C Govindasamy & Dr G Nagarajan** and is submitted in partial fulfilment of the requirements for the current semester of the **B.E Computer Science and Engineering** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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ABSTRACT

The increasing demand for sustainable and efficient urban transportation has led to the widespread adoption of bike-sharing systems in modern cities. Efficient management of these systems requires accurate prediction of bike rental demand, which is influenced by multiple factors such as weather conditions, time of day, season, and working or non-working days. This project focuses on predicting bike-sharing demand using machine learning techniques applied to the Seoul Bike Sharing dataset. The dataset is pre-processed through data cleaning, feature selection, and feature engineering to ensure model reliability and accuracy. Exploratory Data Analysis is conducted to identify patterns and relationships among key variables affecting bike usage. A Random Forest Regression model is developed using Python to forecast hourly bike rental demand, as it effectively captures non-linear relationships within the data. The model performance is evaluated using standard metrics such as Root Mean Square Error (RMSE) and R^2 score to validate prediction accuracy. Furthermore, Python scripting is integrated with Power BI to visualize actual and predicted demand through interactive dashboards, enabling effective interpretation of results. The proposed system supports data-driven decision-making, improves operational efficiency, optimizes resource allocation, and enhances customer satisfaction. This study demonstrates the practical application of machine learning in addressing real-world urban mobility challenges and contributes to the development of sustainable transportation solutions.

KEYWORDS

Machine Learning, Demand Prediction, Random Forest Regression, Python, Power BI, Urban Mobility

TABLE OF CONTENTS

S.NO	TITLE	PAGE NO
1	Abstract	i
2	Acknowledgement	ii
3	Chapter 1: Introduction	4
	1.1 Background Information	4
	1.2 Project Objectives	5
	1.3 Significance of the Study	5
	1.4 Scope of the Study	6
	1.5 Methodology Overview	6
4	Chapter 2: Problem Identification and Analysis	7
	2.1 Description of the Problem	7
	2.2 Evidence of the Problem	7
	2.3 Significance of the Study	8
	2.4 Supporting Data/Research	8
5	Chapter 3: Solution Design and Implementation	9
	3.1 Development and Design Process	9

	3.2 Tools and Technologies Used	9
	3.3 Solution Overview	10
	3.4 Engineering Standards Applied	10
	3.5 Solution Justification	11
6	Chapter 4: Results and Recommendations	12
	4.1 Evaluation of Results	12
	4.2 Challenges Encountered	13
	4.3 Possible Improvements	13
	4.4 Recommendations	13
7	Chapter 5: Reflection on Learning and Personal Development	14
	5.1 Key Learning Outcomes	15
	5.2 Challenges Encountered and Overcome	15
	5.3 Application of Engineering Standards	15
	5.4 Insights into the Industry	16
	5.5 Conclusion of Personal Development	16
8	Chapter 6: Conclusion	17
9	References	18

CHAPTER 1

INTRODUCTION

Urban transportation systems are undergoing a major transformation due to increasing population density, traffic congestion, and environmental concerns. Sustainable and efficient mobility solutions are essential to reduce carbon emissions and improve the quality of urban life. Bike-sharing systems have emerged as an eco-friendly and cost-effective mode of transportation, providing users with flexible mobility options for short-distance travel. However, managing bike availability and predicting rental demand remain significant challenges for service providers. Accurate demand forecasting is crucial for optimizing operations, improving customer satisfaction, and supporting sustainable urban transportation planning.

Advancements in data science and machine learning have enabled the analysis of large-scale transportation data to uncover hidden patterns and trends. By leveraging historical bike rental data along with weather and temporal variables, machine learning models can effectively predict future demand. This project applies machine learning techniques to forecast bike-sharing demand using real-world data, providing insights that support data-driven decision-making and operational efficiency.

1.1 Background Information

Bike-sharing systems have gained widespread adoption in metropolitan cities worldwide as a response to increasing traffic congestion and environmental pollution. These systems allow users to rent bicycles for short durations, promoting last-mile connectivity and reducing dependence on motorized vehicles. The demand for bike rentals is influenced by several factors, including weather conditions, time of day, season, and working or non-working days. Traditional forecasting methods often fail to capture the complex, non-linear relationships between these factors. Machine learning approaches offer a more accurate and scalable solution for demand prediction by learning from historical data patterns.

1.2 Project Objectives

The primary objectives of this project are as follows:

- To analyse historical bike-sharing data and identify key factors influencing rental demand.
- To preprocess and transform raw data for effective model training.
- To develop a machine learning model using Random Forest Regression for predicting bike rental demand.
- To evaluate the performance of the model using appropriate metrics.
- To visualize predicted and actual demand using Python integrated with Power BI.
- To support efficient planning and decision-making for bike-sharing operations.

1.3 Significance of the Study

Accurate prediction of bike-sharing demand is essential for optimizing bike distribution and minimizing operational costs. This study demonstrates how machine learning techniques can be applied to real-world transportation problems to improve service quality and resource utilization. The findings of this project can help bike-sharing operators reduce bike shortages and over-supply issues, enhance customer satisfaction, and promote sustainable urban mobility. Additionally, the project serves as a practical application of data science concepts, bridging the gap between theoretical learning and real-world implementation.

1.4 Scope of the Study

The scope of this project is limited to predicting hourly bike rental demand using the Seoul Bike Sharing dataset. The study focuses on analysing weather-related and temporal features such as temperature, humidity, season, and time of day. The implementation utilizes the Random Forest Regression algorithm and does not include real-time data integration. While the model provides accurate demand predictions, it is intended for academic and analytical purposes rather than full-scale commercial deployment.

1.5 Methodology Overview

The methodology adopted in this project begins with data collection and preprocessing, including handling missing values and selecting relevant features. Exploratory Data Analysis (EDA) is performed to identify trends and patterns in bike rental behaviour. Feature engineering techniques are applied to extract meaningful attributes from date and time data. A Random Forest Regression model is then developed and trained using Python. The model's performance is evaluated using standard metrics such as RMSE and R^2 score. Finally, Python scripting is integrated with Power BI to visualize the results through interactive dashboards, enabling effective interpretation of the predicted bike-sharing demand.

CHAPTER 2

PROBLEM DEFINITION AND ANALYSIS

Bike-sharing systems play an important role in modern urban transportation by offering a sustainable and flexible mobility option. However, managing these systems efficiently is challenging due to fluctuating user demand influenced by multiple factors. Inaccurate demand prediction can lead to issues such as bike shortages at peak hours or excess idle bikes during low-demand periods. These inefficiencies increase operational costs and reduce user satisfaction. Therefore, there is a critical need for an effective and reliable method to predict bike-sharing demand using data-driven approaches.

2.1 Description of the Problem

The primary problem addressed in this project is the inability of traditional planning methods to accurately predict bike-sharing demand under varying conditions. Demand for bike rentals changes significantly based on weather conditions, time of day, season, and whether the day is a working day or holiday. Without accurate forecasting, bike-sharing operators struggle to maintain optimal bike availability across different locations and time periods. This leads to poor resource utilization, increased operational expenses, and dissatisfaction among users due to unavailability or overcrowding of bikes.

2.2 Evidence of the Problem

Historical bike rental data reveals significant variations in demand across different time intervals and environmental conditions. Peak demand is commonly observed during morning and evening commuting hours, while demand drops during late-night and adverse weather conditions such as heavy rain or snow. Additionally, weekends and holidays show different usage patterns compared to working days. These fluctuations highlight the limitations of manual estimation and rule-based approaches, emphasizing the need for a predictive system that can adapt to complex and dynamic patterns in bike-sharing usage.

2.3 Data Analysis and Observations

Exploratory Data Analysis (EDA) was conducted on the Seoul Bike Sharing dataset to identify trends and relationships among various features. The analysis showed that temperature and hour of the day have a strong influence on bike rental demand. Higher demand was observed during moderate weather conditions and peak commuting hours. Conversely, extreme weather conditions and low visibility were associated with reduced bike usage. Seasonal trends also indicated higher demand during spring and summer months. These observations confirm the presence of non-linear relationships in the data, making machine learning techniques suitable for accurate demand prediction.

2.4 Supporting Data and Research

Several recent studies support the use of machine learning models for predicting bike-sharing demand. Research has shown that ensemble learning techniques, such as Random Forest Regression, outperform traditional statistical methods by effectively capturing complex relationships between multiple influencing factors. Studies using real-world bike-sharing datasets demonstrate improved forecasting accuracy when weather and temporal features are incorporated into predictive models. These findings validate the approach adopted in this project and highlight the effectiveness of machine learning in addressing real-world urban mobility challenges.

CHAPTER 3

SOLUTION DESIGN AND IMPLEMENTATION

To address the challenges identified in demand forecasting for bike-sharing systems, a structured and data-driven solution is designed and implemented in this project. The proposed solution utilizes machine learning techniques to analyse historical bike rental data along with weather and temporal variables. The system is developed to accurately predict hourly bike rental demand and provide meaningful insights through interactive visualizations. The design focuses on scalability, accuracy, and practical applicability for operational planning and decision-making.

3.1 Development and Design Process

The development process follows a systematic approach beginning with data acquisition and preprocessing. The Seoul Bike Sharing dataset is collected and examined for missing values, inconsistencies, and outliers. Data preprocessing techniques such as data cleaning, feature selection, and feature engineering are applied to prepare the dataset for model training. Exploratory Data Analysis is conducted to understand demand patterns and relationships between variables. Based on the analysis, a Random Forest Regression model is selected due to its ability to handle non-linear relationships and improve prediction accuracy. The model is trained and tested using split datasets to ensure reliability and robustness.

3.2 Tools and Technologies Used

The implementation of the proposed solution involves the use of the following tools and technologies:

- Python: Used for data preprocessing, model development, and evaluation.

- Pandas and NumPy: Employed for data manipulation and numerical computations.
- Scikit-learn: Utilized for implementing the Random Forest Regression algorithm and performance evaluation.
- Matplotlib and Seaborn: Used for data visualization during exploratory analysis.
- Power BI: Integrated with Python scripts to create interactive dashboards for visualizing actual and predicted bike demand.
- Jupyter Notebook: Used as the development environment for experimentation and analysis.

3.3 Solution Overview

The proposed solution consists of a machine learning-based prediction system that forecasts hourly bike-sharing demand. The system takes processed historical data as input and uses weather and temporal features to generate predictions. The trained Random Forest model produces accurate demand estimates, which are then visualized using Power BI dashboards. These dashboards enable stakeholders to compare predicted and actual demand, identify peak usage periods, and make informed decisions regarding bike distribution and operational planning.

3.4 Engineering Standards Applied

The project follows standard engineering practices to ensure quality and reliability. Data preprocessing adheres to data quality standards by handling missing values and outliers effectively. Model development follows machine learning best practices, including proper train-test splitting, feature scaling where necessary, and performance evaluation using standard metrics such as RMSE and R^2 score. Code modularity, documentation, and reproducibility are

maintained throughout the development process. Visualization standards are applied in Power BI to ensure clarity, accuracy, and ease of interpretation.

3.5 Solution Justification

The selection of a Random Forest Regression model is justified due to its strong performance in handling complex and non-linear datasets. Unlike simple regression models, Random Forest reduces overfitting by combining multiple decision trees, resulting in improved prediction accuracy. The integration of Python with Power BI enhances the practical value of the solution by providing real-time-like visualization and user-friendly dashboards. Overall, the proposed solution effectively addresses the identified problem by delivering accurate demand forecasts, supporting efficient resource allocation, and improving the operational performance of bike-sharing systems.

CHAPTER 4

RESULTS AND RECOMMENDATIONS

This chapter presents the results obtained from the machine learning model developed for bike-sharing demand prediction and evaluates the overall performance of the system. The outcomes of the model are analysed using appropriate performance metrics to understand its accuracy, reliability, and effectiveness in predicting bike demand under different conditions. The results highlight how various factors such as time, weather conditions, and user patterns influence bike usage.

In addition to presenting the results, this chapter discusses the challenges encountered during data preprocessing, model selection, and training phases. Issues such as missing data, feature selection, and model overfitting are examined, along with the steps taken to address them. The limitations of the current approach are also identified to provide a realistic assessment of the model's capabilities.

Based on the analysis and experimental results, possible improvements and practical recommendations are proposed to enhance system performance and real-world applicability. These include the use of advanced machine learning algorithms, incorporation of additional real-time data sources, and optimization techniques to improve prediction accuracy. The recommendations aim to guide future work and support the development of a more robust and scalable bike-sharing demand prediction system.

4.1 Evaluation of Results

The performance of the Random Forest Regression model was evaluated using standard metrics such as Root Mean Square Error (RMSE) and R^2 score. The results indicate that the model effectively captures the relationship between weather, temporal features, and bike rental demand. Predicted values closely follow actual demand trends, especially during peak hours. The integration of Python with Power BI enabled clear visualization of predicted versus actual

bike rental counts, allowing easy interpretation of results. Overall, the model demonstrated high accuracy and reliability in forecasting hourly bike-sharing demand.

4.2 Challenges Encountered

Several challenges were encountered during the development and implementation of the project. Data preprocessing required careful handling of missing values and outliers to avoid biased predictions. Feature selection was critical, as irrelevant features could reduce model performance. Integrating Python scripts within Power BI also required proper configuration and compatibility handling. Additionally, variations in weather conditions introduced complexity in accurately modelling demand patterns. Despite these challenges, appropriate preprocessing techniques and model tuning helped overcome these issues.

4.3 Possible Improvements

Although the proposed system delivers satisfactory results, there is scope for improvement. Incorporating real-time weather data can enhance prediction accuracy by adapting to current conditions. Advanced machine learning algorithms such as XGBoost or neural networks may further improve performance. Hyperparameter optimization techniques can be applied to fine-tune the model. Additionally, incorporating location-specific data for individual bike stations can enable more granular and precise demand forecasting.

4.4 Recommendations

Based on the results of this study, it is recommended that bike-sharing operators adopt machine learning-based demand prediction systems for effective planning and resource management. Regular model retraining using updated data can maintain accuracy over time. Integrating predictive models with operational dashboards can support real-time decision-making. Furthermore, expanding the system to include real-time data sources and advanced analytics can significantly enhance the efficiency and sustainability of bike-sharing services.

CHAPTER 5

REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

This project served as an important learning platform that significantly enhanced both technical competence and personal development. Working on a machine learning–based bike-sharing demand prediction system provided valuable hands-on experience in applying theoretical concepts to solve real-world problems using data-driven approaches. It enabled a deeper understanding of how machine learning models are designed, trained, evaluated, and optimized in practical scenarios.

Through this project, strong technical skills were developed in areas such as data collection, data preprocessing, feature selection, and model implementation. The use of programming tools and machine learning libraries improved coding proficiency and problem-solving ability. Analysing large datasets and interpreting model outputs strengthened analytical thinking and decision-making skills, which are essential for engineering and industry-level applications.

In addition to technical learning, the project contributed significantly to personal and professional development. It encouraged effective time management, systematic planning, and perseverance in overcoming challenges such as data inconsistencies and model limitations. Working through these challenges enhanced self-confidence and adaptability. The project also fostered a sense of responsibility and professionalism by emphasizing documentation, result interpretation, and clear presentation of findings.

Overall, this project successfully bridged the gap between academic theory and industry-oriented application. The knowledge and experience gained have laid a strong foundation for future learning and career growth in the field of data science and machine learning, while also cultivating essential soft skills required for professional success.

5.1 Key Learning Outcomes

The project enabled the acquisition of strong technical skills in data analysis and machine learning. Key learning outcomes include understanding data preprocessing techniques such as handling missing values, feature engineering, and exploratory data analysis. Practical experience was gained in implementing Random Forest Regression for predictive modeling and evaluating model performance using standard metrics. The project also improved proficiency in Python programming and the use of data science libraries. Additionally, integrating Python with Power BI strengthened skills in data visualization and effective presentation of analytical results.

5.2 Challenges Encountered and Overcome

Several challenges were encountered during the project execution. Managing large datasets and ensuring data quality required careful preprocessing and validation. Selecting relevant features and tuning model parameters was initially complex but was resolved through iterative experimentation. Technical difficulties in integrating Python scripts with Power BI were addressed through systematic debugging and testing. Overcoming these challenges improved problem-solving abilities, patience, and confidence in handling complex engineering tasks.

5.3 Application of Engineering Standards

Engineering standards were applied throughout the project to ensure quality, accuracy, and reliability. Standard practices in data handling, model validation, and performance evaluation were followed. Ethical considerations related to data usage and result interpretation were maintained. The project also emphasized code modularity, documentation, and reproducibility, aligning with professional engineering and software development standards.

5.4 Insights into the Industry

The project provided valuable insights into how machine learning is used in the transportation and urban mobility industry. It highlighted the importance of predictive analytics in optimizing operations and improving service quality. The study demonstrated how data-driven solutions support sustainable development and informed decision-making. Exposure to industry-relevant tools such as Power BI and machine learning frameworks enhanced understanding of current trends and expectations in the data analytics domain.

5.5 Conclusion of Personal Development

Overall, this project significantly contributed to personal and professional growth. It enhanced technical knowledge, analytical thinking, and problem-solving skills. The experience strengthened confidence in applying machine learning techniques to real-world problems and reinforced the importance of continuous learning. The knowledge and skills gained through this project will serve as a strong foundation for future academic and professional endeavours.

CHAPTER 6

CONCLUSION

This project successfully demonstrated the practical application of machine learning techniques for predicting bike-sharing demand using the Seoul Bike Sharing dataset. By systematically analysing key weather-related and temporal factors such as temperature, humidity, season, rainfall, snowfall, and hour of the day, a robust and effective prediction model was developed using Random Forest Regression. These factors were found to have a significant impact on bike rental behaviour, enabling the model to capture complex, non-linear relationships within the data.

The integration of Python for model development and Power BI for data visualization played a crucial role in enhancing the interpretability of the results. Visual comparisons between actual and predicted bike rental demand provided clear insights into model performance and facilitated informed decision-making. This combined analytical and visualization approach improved the overall understanding of demand patterns and trends.

The results clearly indicate that machine learning-based models offer superior demand forecasting accuracy compared to traditional estimation and manual forecasting methods. The proposed system can assist bike-sharing service providers in optimizing bike distribution, improving resource allocation, reducing operational costs, and minimizing bike shortages or surpluses. Accurate demand prediction also contributes to enhanced customer satisfaction by ensuring better availability and service reliability.

Overall, this project highlights the importance of data-driven approaches in addressing real-world urban mobility challenges. It demonstrates how machine learning can support the development of sustainable, efficient, and intelligent bike-sharing systems. The insights gained from this work can be extended to future research and real-time implementations, contributing to smarter transportation planning and sustainable urban development.

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